

MCALESTER, OK, USA

WMO: 723566

Lat: 34.882N

Lon: 95.783W

Elev: 770

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	20.5	4.5	7.1	21.9	9.1	9.0	25.4	24.9	49.6	22.4	48.3	7.2	0	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	100.0	75.0	96.9	75.1	93.9	75.3	79.1	91.3	78.1	90.3	77.3	89.3	8.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	138.2	85.5	74.7	133.7	84.5	73.6	128.9	83.3	43.1	91.4	42.1	90.5	41.1	89.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	18.8	17.0	8.1	102.7	5.0	3.9	4.5	105.6	1.6	107.9	-1.3	110.1	-4.9	112.9
			7.1	81.2	4.8	1.7	3.7	82.4	0.9	83.4	-1.8	84.3	-5.2	85.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	41.4	45.4	53.9	62.0	70.0	78.2	82.5	82.1	74.6	63.6	52.6	43.6	
	DBStd	16.74	10.90	11.13	10.12	8.21	7.00	4.95	4.84	5.16	7.23	8.86	9.64	10.03	
	HDD50	936	309	200	76	9	1	0	0	0	0	7	83	251	
	HDD65	3032	733	551	364	151	34	0	0	0	12	137	383	667	
	CDD50	5527	41	72	197	368	622	847	1006	995	739	427	162	51	
	CDD65	2148	1	3	20	59	190	397	541	530	301	92	12	2	
	CDH74	22972	3	28	140	453	1578	4105	6499	6258	3048	786	70	4	
	CDH80	10579	0	7	20	73	428	1750	3392	3313	1364	223	9	0	
Wind	WSAvg	7.2	8.2	8.4	9.0	8.9	7.4	6.4	5.8	5.4	5.5	6.6	7.6	7.7	
Precipitation	PrecAvg	43.9	2.0	2.6	3.8	4.1	5.7	4.4	2.4	2.9	4.8	4.4	3.4	2.9	
	PrecMax	66.4	6.3	7.3	8.9	10.7	12.5	8.9	6.5	9.5	12.5	17.4	9.8	8.3	
	PrecMin	20.7	0.0	0.3	0.9	0.2	0.8	0.5	0.0	0.0	0.3	0.1	0.5	0.2	
	PrecStd	11.5	1.5	1.7	2.1	2.4	3.0	2.2	1.9	2.2	3.2	4.1	2.3	2.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	77.2	82.7	85.8	90.7	96.7	103.8	104.2	99.8	89.6	80.8	73.5	
		MCWB	57.4	59.8	64.7	66.4	74.2	75.7	74.3	74.4	73.2	70.6	65.1	62.8	
	2%	DB	68.4	72.0	77.8	81.9	87.4	93.4	100.0	100.4	94.6	85.9	75.5	69.1	
		MCWB	58.0	58.9	62.4	66.2	72.9	76.1	75.3	74.9	73.4	68.6	62.3	59.3	
	5%	DB	64.1	67.8	74.1	79.2	84.5	91.1	96.7	97.4	91.2	82.1	71.9	64.5	
		MCWB	55.7	57.7	60.9	64.6	71.5	75.4	76.0	74.9	72.6	67.3	61.8	56.5	
	10%	DB	60.1	63.7	70.4	76.2	82.0	88.8	93.4	94.0	87.9	78.4	68.4	60.5	
		MCWB	50.9	55.2	58.9	63.3	70.6	75.0	76.2	74.9	71.8	66.1	59.4	52.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	65.4	67.7	71.5	77.3	80.5	80.5	79.8	77.6	74.4	69.4	65.8
			MCDB	68.7	71.6	78.4	79.8	87.1	91.2	92.3	92.1	90.2	83.3	76.2	71.7
2%		WB	61.1	63.1	65.3	69.5	75.3	78.5	79.3	78.4	76.1	72.6	66.2	62.1	
		MCDB	66.0	68.1	73.8	77.5	83.8	89.3	91.8	91.4	88.1	80.9	71.9	67.4	
5%		WB	57.0	59.6	63.4	68.1	73.8	77.4	78.3	77.4	74.9	70.8	63.7	58.6	
		MCDB	62.3	66.4	71.2	75.5	82.0	88.0	90.7	90.6	86.1	77.8	69.5	63.4	
10%		WB	51.8	55.4	60.9	66.3	72.3	76.2	77.4	76.6	73.7	68.7	61.0	54.0	
		MCDB	58.5	62.5	69.0	73.8	80.2	85.9	89.6	89.3	83.8	75.6	67.6	58.8	
Mean Daily Temperature Range	5% DB	MDBR	21.0	20.8	21.3	21.7	18.8	18.5	19.9	21.0	21.7	22.1	21.2	19.6	
		MCDBR	26.7	26.2	24.7	23.6	20.5	20.4	23.8	25.4	24.8	25.0	23.8	24.8	
		MCWBR	17.2	16.7	13.6	12.2	9.9	8.1	7.0	7.3	8.7	11.5	14.1	16.9	
	5% WB	MCDBR	21.0	22.2	20.7	18.1	17.3	18.1	19.6	20.3	19.8	18.0	20.4	20.4	
		MCWBR	16.9	16.8	13.7	12.1	9.5	8.5	7.7	7.4	7.7	10.1	15.4	16.6	
Clear-Sky Solar Irradiance	taub		0.307	0.315	0.349	0.384	0.417	0.427	0.443	0.441	0.394	0.348	0.322	0.301	
	taud		2.494	2.493	2.422	2.332	2.302	2.307	2.306	2.284	2.377	2.493	2.510	2.525	
	Ebn at Noon		286	296	293	286	276	272	267	266	274	280	278	282	
	Edh at Noon		27	30	35	40	42	41	41	41	36	30	26	24	
All-Sky Solar Radiation	RadAvg		834	991	1326	1630	1785	2041	2075	1898	1581	1205	887	715	
	RadStd		87	147	89	113	174	225	125	122	145	171	110	94	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+166
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137

Nomenclature: See separate page